

F1

$$\omega = \omega_0 + \alpha t$$

$$\omega = (2)(2) = 4 \left[\frac{\text{rad}}{\text{s}} \right]$$

$$\Rightarrow v = R\omega = (0,3)(4)$$

$$v = 1,2 \left[\frac{\text{m}}{\text{s}} \right]$$

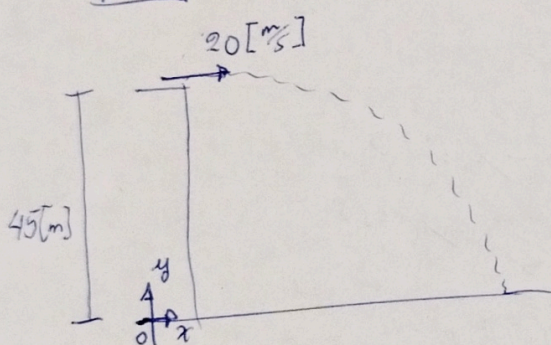
$$\Rightarrow a_n = \frac{v^2}{R} = \frac{1,2^2}{0,3} = 4,8 \left[\frac{\text{m}}{\text{s}^2} \right]$$

$$\Rightarrow a_t = R\alpha = (0,3)(2) = 0,6 \left[\frac{\text{m}}{\text{s}^2} \right]$$

$$\Rightarrow a = \sqrt{a_n^2 + a_t^2}$$

$$a = 4,84 \left[\frac{\text{m}}{\text{s}^2} \right]$$

F2

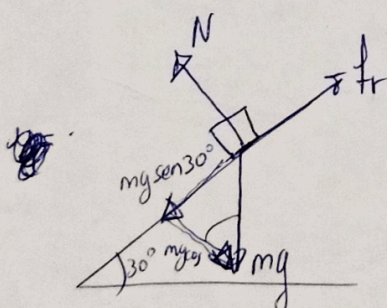


$$y = 45 - \frac{1}{2} g t^2$$

$$\frac{1}{2} g t^2 = 45$$

$$t = \sqrt{\frac{90}{g}} = 3 \left[\text{s} \right]$$

F3



$$g = 9,8 \left[\frac{\text{m}}{\text{s}^2} \right]$$

$$\left\{ \begin{array}{l} m g \sin 30^\circ - f_r = m a \\ N - m g \cos 30^\circ = 0 \rightarrow N = m g \cos 30^\circ \end{array} \right.$$

$$m g \sin 30^\circ - \mu N = m a$$

$$m g \sin 30^\circ - \mu m g \cos 30^\circ = m a$$

$$a = g \sin 30^\circ - \mu g \cos 30^\circ$$

$$a = 2,35 \left[\frac{\text{m}}{\text{s}^2} \right]$$

F4

$$y = 1,25 - \frac{1}{2} g t^2$$

$$0 = 1,25 - \frac{1}{2} g t^2$$

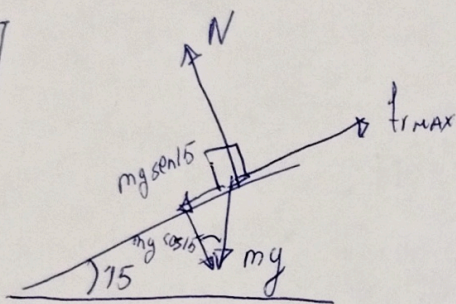
$$t = \sqrt{\frac{2,5}{g}}$$

$$\theta = \theta_0 + \omega t$$

$$2\pi = \omega \sqrt{\frac{2,5}{g}}$$

$$\omega = 4\pi \left[\frac{\text{rad}}{s} \right]$$

F5



$$mg \sin 15^\circ - f_{\text{max}} = 0$$

$$N - mg \cos 15^\circ = 0 \rightarrow N = mg \cos 15^\circ$$

$$mg \sin 15^\circ - \mu N = 0$$

$$mg \sin 15^\circ = \mu mg \cos 15^\circ$$

$$\frac{\sin 15^\circ}{\cos 15^\circ} = \mu$$

$$\mu = \tan 15^\circ$$

$$\mu = 0,268$$